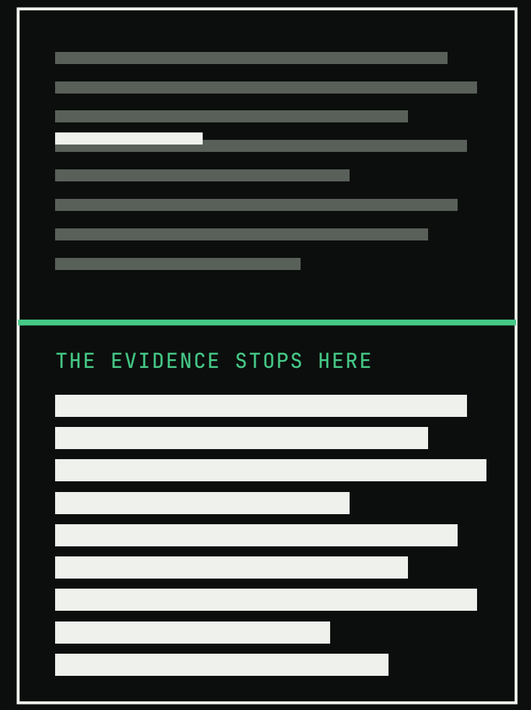
OPEN AGENT HACKATHON 2026 · TRACK 1

Sakaime 境目

Know where the evidence stops.



A hospital team asks whether the evidence behind a vendor's AI claim covers the exact version, patients, setting, purpose and date it has in mind. Sakaime finds the sources, joins them, and answers no more broadly than the evidence allows.

Zetaris · Meterless · NVIDIA Nemotron via Nightshift · Solo build: Prateek · Apache-2.0

Vendor claims and the hospital requirement are constructed. Regulator, registry and literature records are real. The product screens in this deck run in sample mode on recordings made on 2026-10-27.

Sources: README.md · metadata.json

THE PROBLEM

A claim can be true and still be wrong for you.

Three constructed vendor claims about one cleared product. Each is checked against the same real evidence.

CONSTRUCTED VENDOR CLAIM EA-04

“The 2026 release, version 3.0, is covered by the same clearance and the same pivotal study.”

SOFTWARE VERSION

Claimed	**3.0**
Evidence	**study: 2.1.0**

✕ Disjoint

— Not enough evidence

CONSTRUCTED VENDOR CLAIM EA-10

“In the population of the pivotal study, extended to include adults aged 18 to 21 (adults aged 18 and older with …”

POPULATION

Claimed	**ages 18 and above**
Evidence	**regulator summary: ages 22 and above**

Partial

Supported only for a narrower use

CONSTRUCTED VENDOR CLAIM EA-03

“Supports prioritisation of diabetic eye disease for patients presenting to the emergency department or admitted to …”

CARE SETTING

Claimed	**emergency department, inpatient ward**
Evidence	**eye care, primary care**

✕ Disjoint

— Not enough evidence

The same product passes elsewhere: the planned use date, 2027-01-15, lies inside an authorization in force since 2020-08-03. Sakaime reports each part of the use on its own.

Sources: output_examples/dossiers/eyeart-k200667.dossier.json · data/sample/run/labels.json · input_examples/cases/eye… constructed claims; recorded run

THE USER AND THE DECISION

A reviewer must decide what a vendor claim covers.

IMPACT, 30 POINTS

“Consistency with the challenge statements: how effectively the solution addresses a meaningful real-world problem for its intended users”

THE USER

Hospital AI-evaluation reviewer

Clinical informatics analyst, procurement analyst or quality lead on an AI intake committee. Knows the hospital's own requirements.

THE DECISION

Take this claim into technical and clinical review for the use we plan, or send it back with a request.

NOT MEASURED

No hospital reviewer or buyer has been interviewed. Demand is not verified; nothing here is evidence of it.

FIVE JOBS THE PRODUCT DOES FOR EACH CLAIM

1 **Does the evidence cover the use we have in mind?**
Shown with the exact passage, page and source.

2 **What is missing?**
Written as a request that can be sent to the vendor.

3 **If we narrow or widen the use, what changes?**
Recomputed in the browser, no model called.

4 **When something we relied on is updated, what does it touch?**
Only the dependent findings are marked stale.

5 **Can I hand the committee a record and re-check it later?**
A signed bundle with an offline verifier.

Sources: docs/01-prd.md · README.md

WHY THE EVIDENCE IS FRAGMENTED

Five systems, five identifiers that rarely cross.

Regulator `K200667`
FDA 510(k) / De Novo number

Registry `NCT03112005`
ClinicalTrials.gov id

Literature `PMID 34779843`
PubMed id or DOI

Vendor packet `EyeArt 2.1.0`
product name and a version string

Hospital `CASE-2026-014`
its own case and policy ids



Twelve real FDA-listed AI devices, probed keyless on 2026-10-27. A tick means the test holds; a cross means the sources collide or conflict. Links are therefore shown as ranked candidates with a reason, never as fact below an exact identifier.

Sources: docs/reports/feasibility-report.md

probe of 12 real devices, 2026-10-27, one reader

WHAT EXISTS TODAY

Lookup tools return evidence, not verdicts.

SAME 26 CLAIMS AND RECORDS	TAKES A CLAIM	VERDICT	KEY RE-DERIVABLE	EXACT LABEL, DEV / HELD-OUT	UNSUPPORTED CALLED SUPPORTED, DEV / HELD-OUT
Constat (free MCP lookup)	No: one K or De Novo number	None	7 of 26	n/a	n/a
HealthIT.com ledger	No: a filter by product	None	0 of 26	n/a	n/a
One Nemotron call, all records	Yes	Yes	n/a	44.9% / 63.3%	39.2% / 22.2%
Tool-using Nemotron agent	Yes	Yes	n/a	46.2% / 60.0%	29.4% / 20.0%
Sakaime	Yes	One of five labels	n/a	34.6% / 45.0%	0.0% / 0.0%

EIGHTEEN PRODUCTS READ

The competition report scores 18 products and frameworks on five capabilities, these two included. None checks a claim against version, population, setting and intended use: 0 read YES.

READ THIS BEFORE QUOTING THE GAP

The Gate 0 rule says the gap stands, and that is weak: Sakaime's exact-label accuracy is below both generic Nemotron arms, one builder wrote the claims and the key, and Constat and HealthIT.com were not run on the held-out claims.

Sources: evals/gate0/COMPETITOR-ARMS.md · docs/reports/competition-report.md · evals/RESULTS.md · +2

builder-labelled; same inputs; Gate 0

WHAT WE BUILT

Claims in. Labelled, sourced, signed answers out.

IN

The hospital's intended use

The vendor's claims

constructed, labelled as such

→

1	**Discover**	Onboard a source nobody wrote an adapter for; refuse unsafe addresses	`src/discovery/`
2	**Join**	One federated SQL over Postgres, registry snapshots and vendor files	`src/federation/`
3	**Resolve**	Typed links; uncertain stays uncertain	`src/identity/`
4	**Read**	Anchored quotes from PDFs and APIs; long documents in chunks	`src/extraction/`
5	**Compile**	Seven dimensions, five labels, smallest mismatch	`src/compiler/`

→

OUT, FOR EVERY CLAIM

One of five labels

The smallest mismatch

What would settle each unknown

A signed dossier

14
screens in the web workstation

3,140
automated tests, offline

157
fault drills, all in the correct state

0
keys needed to replay the recorded run

Sources: README.md · docs/05-requirements-matrix.md

HOW IT WORKS, END TO END · EYEART K200667

One real claim: question, sources, join.

1 Question

The hospital states its intended use. Ten vendor claims are checked against the same evidence.

INTENDED USE OF THE HOSPITAL, CONSTRUCTED

“Can our primary-care clinics use EyeArt 2.1.0 to screen adults with diabetes and no previous diagnosis of more than mild diabetic retinopathy, and does the evidence behind a sensitivity of at least 85 percent apply to that use?”

`10 constructed vendor claims`

`sensitivity at least 85 percent`

`Review desk`

2 Discover

The guard reads the request, then the planner picks sources from the registered cards.

9 SOURCE CARDS REGISTERED; 6 USED

Regulator — 3 used
openFDA 510(k), openFDA UDI, FDA decision summary (PDF)

Registry — 1 used
ClinicalTrials.gov v2

Literature — 1 used
PubMed E-utilities

Hospital — 1 used
hospital policy database (Postgres)

`src/discovery/` `src/sources/`

3 Join

One federated SQL statement over three kinds of silo. Zetaris runs the join, not our code.

```
SELECT
  c.case_id,
  c.intended_purpose,
  c.intended_setting,
  CAST(c.intended_min_age AS INT)
AS intended_min_age,
  s.product_name,
  vc.claim_id,
  vc.claimed_version,
  CAST(vc.claimed_min_age_years AS INT)
AS claimed_min_age_years,
  ...
```

`3 statements` `10 + 10 + 3 rows` `recorded answers`

`src/federation/` `Evidence matrix`

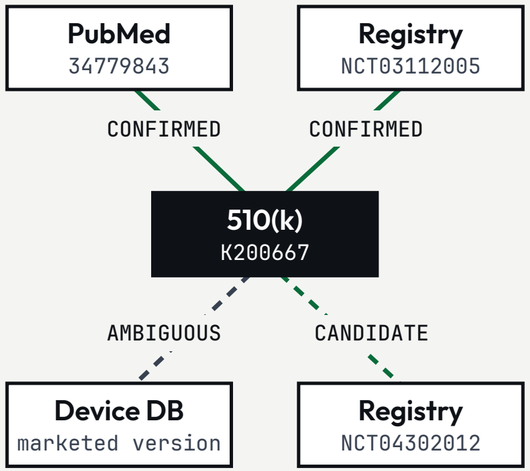
Query id of the first statement: `2a05328a-7171-48c2-b8c8-8a46bf1c0634`. The join answer equals an independent DuckDB run cell by cell on the real engine (gate Z-HEAD).

HOW IT WORKS, END TO END · EYEART K200667

Identity, quote, compile.

4 Resolve identity

6 typed links: 4 confirmed, 1 candidate, 1 ambiguous. A score alone never confirms.



The candidate scores 4.325 bits: the version is stated on one side only.

src/identity/ · Claim X-ray

5 Extract

175 assertions, each anchored to a page. Code checks the quote exists; a separate check asks whether it supports the claim.

FDA DECISION SUMMARY K200667, PAGE 7

> The target population of the EyeArt pivotal clinical trial analysis was asymptomatic persons aged 22 years and older who were diagnosed with diabetes and not diagnosed with more than mild diabetic retinopathy.

101 anchors · quote exists: code · supports claim: separate

src/extraction/

6 Compile

Seven dimensions, each contained, partial, disjoint or unknown. Shown for the claim about version 3.0.

Product	K200667	✓ Contained
Version	3.0	✕ Disjoint
Population	ages 22 and abov…	◐ Partial
Setting	eye care, primar…	✓ Contained
Purpose	detection aid	✓ Contained
Timing		– Unknown
Metric		– Unknown

src/compiler/ · Applicability lab

HOW IT WORKS, END TO END · EYEART K200667

Label, break it on purpose, sign it.

7 Label

A fixed table turns the dimensions into one of five labels, the smallest mismatch and a request.

Not enough evidence

SMALLEST MISMATCH

Version: claimed 3.0, evidence 2.1.0

TO SETTLE IT

From the vendor: study results for software version 3.0, or a written change-control statement saying ...

`1 × supported` `3 × narrower` `1 × contradicted` `2 × not enough` `3 × review`

`src/compiler/` `Claim X-ray`

8 Drill

Switch the study source off through the real code and run the case again.

CLAIM EA-01, THE ONE SUPPORTED CLAIM

✓ Supported for this use

↓

Not enough evidence

11 findings relied on the lost source and are marked stale. Nothing became supported because something was lost.

`invariant held`

`src/service/drills.ts` `Resilience`

9 Dossier

The whole run, signed, with a verifier that recomputes every hash and reruns the compiler.

BUNDLE ROOT (SHA-256)

7df8800d3f3adbae ...

`Ed25519` `demo key` `21 sections`

Change one character and the verifier names the failing item. A signature shows the bundle is unchanged, not that any finding is right.

`src/dossier/` `Dossier`

ARCHITECTURE

Nine layers. Code decides; Nemotron proposes.

> TECHNICAL, 20 POINTS
> “How well is the solution built, including its technical implementation, reliability and architecture?”

L0	**Scout guard**	intent, injection quarantine, signed export contracts	Meterless Scout Intent
L1	**Source discovery**	allowlist, SSRF-safe fetch, bounded probe, schema fingerprint, semantic validation, review, registration	Zetaris catalog · Nemotron mapping
L2	**Zetaris federation**	one SQL over Postgres, registry snapshots and vendor files; policies; Unified Semantic Layer	Zetaris
L3	**Identity resolution**	typed links; an uncertain link stays uncertain	Nemotron in the uncertain band
L4	**Extraction**	anchored PDF and API parsing; quote-exists is separate from semantic support	Meterless Markovian · Nemotron extractor
L5	**Evidence memory**	bi-temporal World Model; H-MEM for confirmed links and precedents	Meterless World Model · Meterless H-MEM
L6	**Applicability compiler**	per-dimension containment, label table, smallest mismatch, counterfactuals	code only: no model
L7	**Investigation loop**	planner, retriever, resolver, extractor, critic, reporter; bounded	Nemotron roles
L8	**Dossier**	signed bundle, offline verifier, lineage events, reviewer decisions	Nightshift receipts

A model proposal never becomes a fact until a deterministic check or a reviewer accepts it.

Sources: README.md · docs/02-architecture.md

design contract; build matches except where the docs say

TECHNICAL STACK

One language, typed at every boundary.

RUNTIME	
TypeScript	Node 22.19+
Zod	v3
Hono	v4
node:sqlite	local state
DuckDB	test oracle only
pdfjs-dist	v6
Signatures	node:crypto

WEB APP	
React	v19
Vite	v7
React Router	v7
Outfit	display and body
JetBrains Mono	ids and SQL

VERIFICATION	
Vitest	v4
fast-check	v3
Playwright	v1
axe	accessibility
3,140 tests	133 files

SPONSOR STACK	
Zetaris	2.4.3.1
Spark	3.5.3 (local)
Postgres	v16 silo
Meterless	4 engines
Nemotron	via Nightshift

openFDA 510(k) and UDI · ClinicalTrials.gov v2 · PubMed · Europe PMC · FDA decision summaries (PDF) — public data, no keys; Apache-2.0

Sources: package.json · docs/05-requirements-matrix.md · infra/zetaris/GATES.md · +1

versions read from package.json

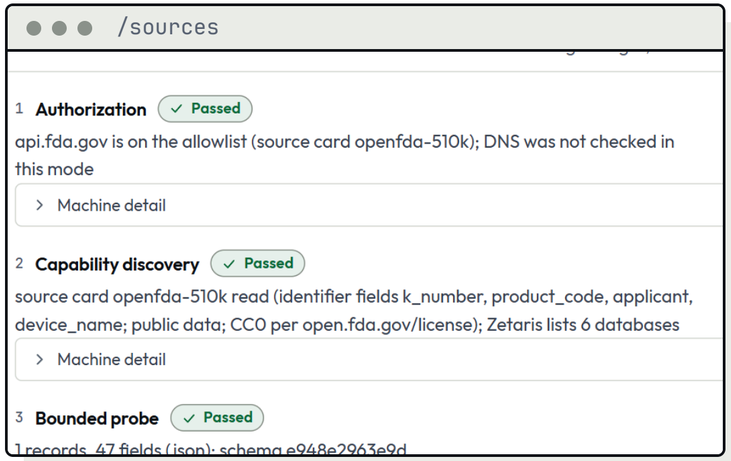
THE TRACK 1 CHALLENGE

Discover, work across, reconcile.

TRACK 1, SOLVING FRAGMENTED INTELLIGENCE

“Demonstrate a clever ability to discover and work across fragmented data sources.”

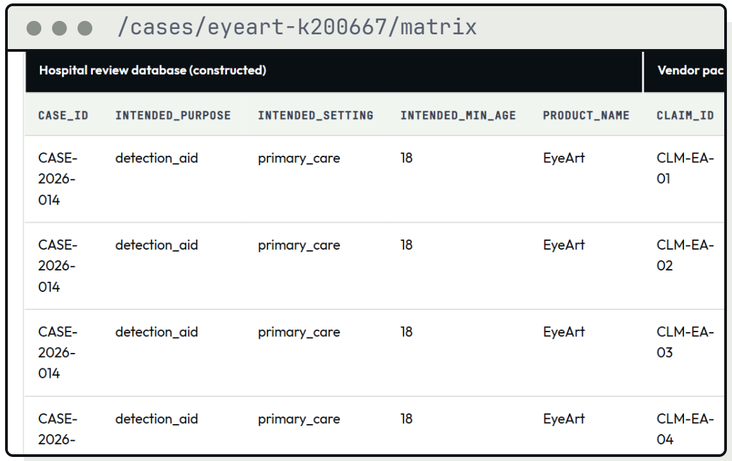
1 Discover



A source nobody wrote an adapter for goes through eight stages. Cloud-metadata and unlisted hosts are refused at stage one.

44/44 must-not-register layouts caught

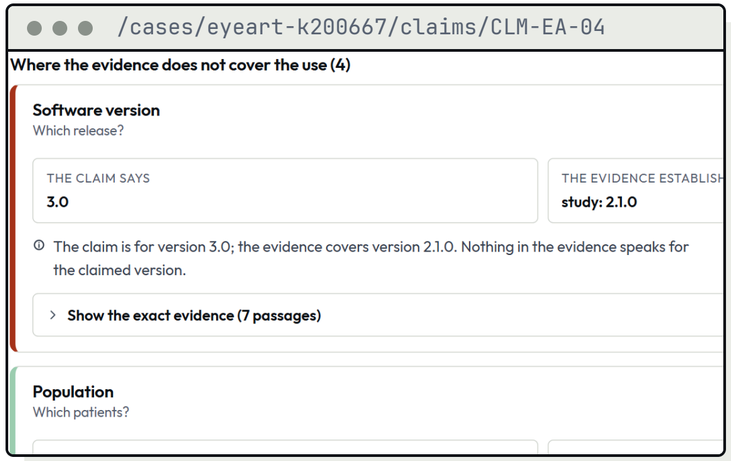
2 Work across



One Zetaris statement joins a hospital database, registry and regulator snapshots and vendor files. Each column names its source.

one statement, 20 rows, equals DuckDB

3 Reconcile



Typed identity links, authority by kind of assertion, and for each dimension the claimed value beside what the evidence establishes.

0 of 60 false merges

The organizers' own good example: “querying three disconnected systems, reconciling conflicting records, and producing a single sourced summary with the systems each fact came from.”

Sources: research/hackathon/organizer/01-welcome-guide.md · research/hackathon/organizer/04-approach-guide-solut… screens: sample mode, recorded replies

WHAT IS NEW

It answers no more broadly than the evidence allows.

INNOVATION, 15 POINTS

“How original or creative is the approach to solving the problem?”

01 Three-valued applicability

Seven dimensions, each contained, partial, disjoint or unknown. Unknown is never read as covered and never as false.

0 of 17 dev and 0 of 15 held-out unsupported claims labelled supported

02 The smallest mismatch, and what settles it

Not a score: the dimensions that stop the claim, both values side by side, and a request written to send to the vendor.

every finding names the passage, page and date

03 Counterfactual lab

Change the version, ages, setting or date and the label recomputes. No model is called, so the same inputs always give the same answer.

slide 19

04 Change-triggered re-evaluation

A World Model records which facts each finding used. When a source changes, only the dependent findings go stale.

9 findings recomputed instead of 36, none missed (constructed session)

05 Onboarding that checks meaning

A mapping is checked for units, scale, identifier family and joins against records already held, not only for types.

0 false accepts of a semantic mutation in 52 that could produce one

THE COST, STATED

The same discipline makes the full pipeline conservative: its exact-label accuracy is below the generic Nemotron arms and it flags well-supported claims it cannot place. Slide 30 shows every row.

Sources: evals/RESULTS.md · docs/02-architecture.md · tests/engines/ABLATIONS.md · +1

compared on the same inputs; see slides 30 and 31

SOURCE DISCOVERY AND SAFE ONBOARDING

New sources are checked for meaning, then registered.

1 **Authorization** 2 **Capability discovery** 3 **Bounded probe** 4 **Mapping proposal** 5 **Semantic check** 6 **Human review** 7 **Registration** 8 **Monitoring**

Layouts that must map, mapped correctly (of 48)



Layouts that must not register, caught (of 44)



102 mutated layouts from four base tables, sealed answers, one labeller. The second run follows fixes made against the same cases, so it is no longer held out. Two known limits are reported, not scored.

21 of 21 SSRF drills correct: private, loopback, link-local, metadata, rebinding, redirects, non-https, oversize, unlisted hosts. Refused at stage one with a reason.

Sources: data/discovery-shift/RESULTS.md · evals/results/discoveryshift.json · evals/results… builder-labelled consistency; 4 base sources; one labeller

IDENTITY RESOLUTION

Link what is exact. Keep the rest uncertain.

FIVE LINK TYPES, EACH ITS OWN CLAIM

SAME_SUBMISSION
one authorization

SAME_PRODUCT
one product line

SAME_MARKETED_VERSION
one software release

STUDY_EVALUATES
a study tested this product

PUBLICATION_REPORTS
a paper reports this study

CONFIRMED CANDIDATE AMBIGUOUS REJECTED

Only an exact identifier or a reviewer confirms. A score cannot.

Study vs. cleared product, 1.742 bits



version differs: 1.7.0 versus 1.7.2

family name shared by 6 reference rows

A real ambiguous link from the Transpara case.

0 of 60
false merges, against 23 of 60 for a fuzzy-name baseline

48 of 48
links called were true (precision)

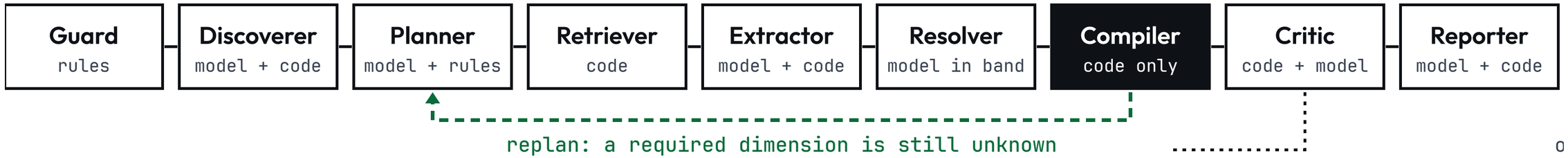
48 of 62
true links found: recall 0.77, the weak number

20 of 22 ambiguous pairs left ambiguous. 144 pairs, 14 families held out in turn, one reader, a development set.

Sources: data/identity-corpus/RESULTS.md · evals/results/identityshift.json · output_examples… builder-labelled development set; one reader; 14 families

AGENT WORKFLOW

A bounded loop that replans and critiques.



REAL TRACE LINES, TRANSPARA CASE (245 LINES ACROSS THREE CASES, 4 REPLANS)

guard request understood as review_claim; the investigation may proceed

planner plan: 2 regulator reads, the decision summary, then 5 searches; cited studies are added once the summary is read

planner replan: reading 4 more candidate records because a required dimension is still unknown

critic the adversarial model raised 5 objections: 0 confirmed by code, 0 unverified, 0 dismissed, 5 already stated

THE ORGANIZERS' NINE BEHAVIOURS

- ✓ Planner delegates sub-tasks
- ✓ Specialist uses data or tools
- ✓ Evaluator critiques the output
- ✗ Agent revises after critique not in the recorded runs
- ✓ A safety role flags unsupported claims
- ✓ Structured messages in shared state
- ✓ Workflow branches on confidence
- ✗ Fallback if a source or model fails not in the recorded runs
- ✓ Final answer carries traceable evidence

Sources: output_examples/traces/transpara-k221347.trace.jsonl · output_examples/traces/eyeart-k200667.trace.js… recorded traces; critic changed no label

THE APPLICABILITY COMPILER

Seven dimensions in, one of five labels out. No model.

COMPARED ONE BY ONE

Product	same product
Version	exact, same release line, unknown
Population	age interval, conditions, exclusions
Setting	ED < hospital < any care setting
Purpose	triage < detection aid < diagnosis
Timing	use date inside the evidence period
Metric	claimed operating point

✓ Contained · ✓ Partial · ✕ Disjoint · — Unknown

A FIXED TABLE DECIDES THE LABEL

✓ Supported for this use
every dimension contained, every required source adequate

✓ Supported only for a narrower use
a narrower use is itself contained

✕ Contradicted by specified evidence
relevant, adequate evidence disagrees

— Not enough evidence
a required dimension or source is unknown or lost

Needs domain review
the rule needs clinical judgement

AUTHORITY DEPENDS ON THE QUESTION

What use was authorized
regulator record

What a study measured
the study report

What this hospital requires
its own policy

What the vendor says
a claim to check

A study cannot widen what was authorized. A later document is not a stronger one.

218 of 222
single-change cases agree with the rule table

24 of 24
positive controls supported

0 of 198
withheld-support cases labelled supported

Consistency, not clinical validity. Concepts are cited from PROBAST+AI, STARD-AI and others; the executable logic is our inference. 17 of 38 registry rules are implemented, 21 are not, each with a reason.

Sources: docs/02-architecture.md · evals/results/scopeshift-a.json · docs/DECISIONS.md · +1 consistency: code against the rule table its authors wrote

WHAT THE PRODUCT REFUSES TO SAY

Five labels. None of them says safe.

Supported for this use

The evidence covers every part of the use.

1 of 10 EyeArt claims

Supported only for a narrower use

It covers a narrower use, which the finding states.

3 of 10 EyeArt claims

Contradicted by specified evidence

Relevant, adequate evidence points the other way.

1 of 10 EyeArt claims

Not enough evidence

A required piece is missing, unreadable or unavailable.

2 of 10 EyeArt claims

Needs domain review

The rule needs clinical judgement the code refuses to make.

3 of 10 EyeArt claims

THE PRODUCT NEVER SAYS

~~safe~~ ~~cleared~~ ~~compliant~~

The label type is closed. The reporter drops any sentence with banned wording. A request to certify a product safe is blocked by the guard.

WHAT IT KEEPS APART

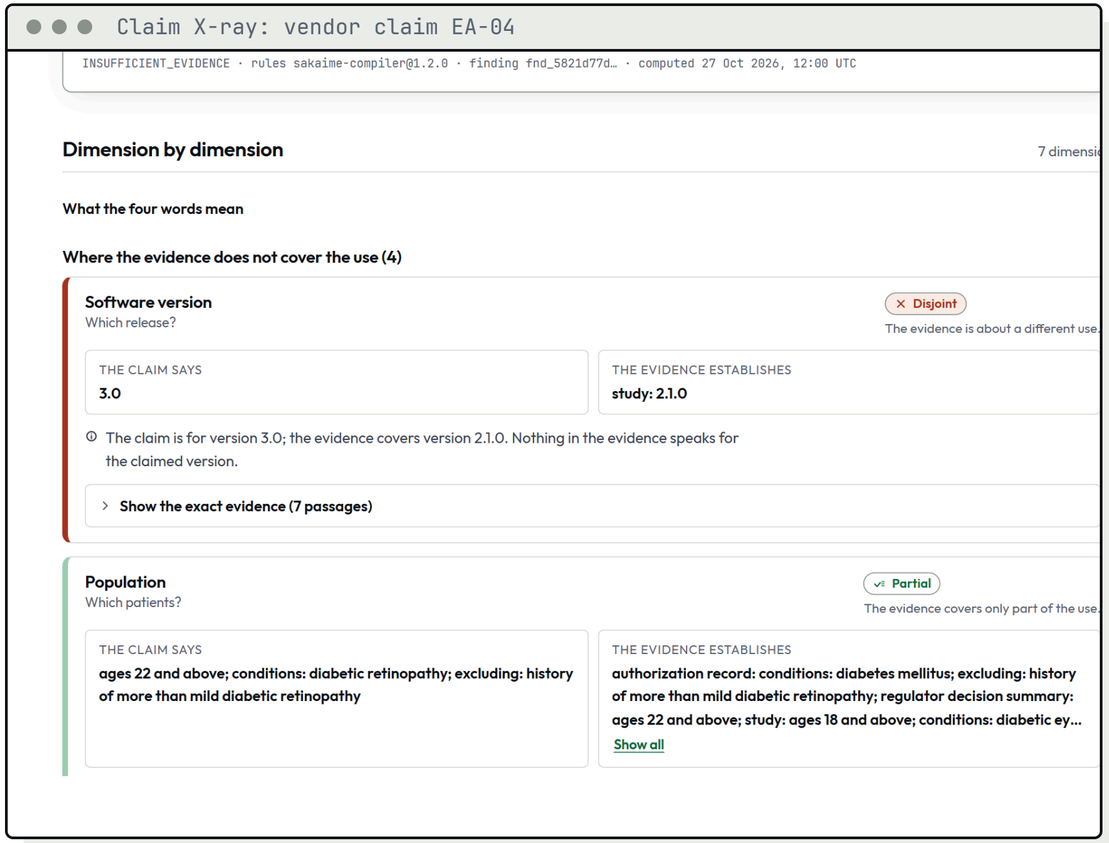
- Unknown is never false and never safe.
- A source that was down is never a source with nothing to say.
- Needs domain review is a refusal to decide, not a middle score.
- The Lunit and Transpara cases stay not enough evidence throughout.

Guard on its own corpus: certify-this requests blocked 25 of 34; the rest are sent to clarification. Held-out wording is weaker (slide 28).

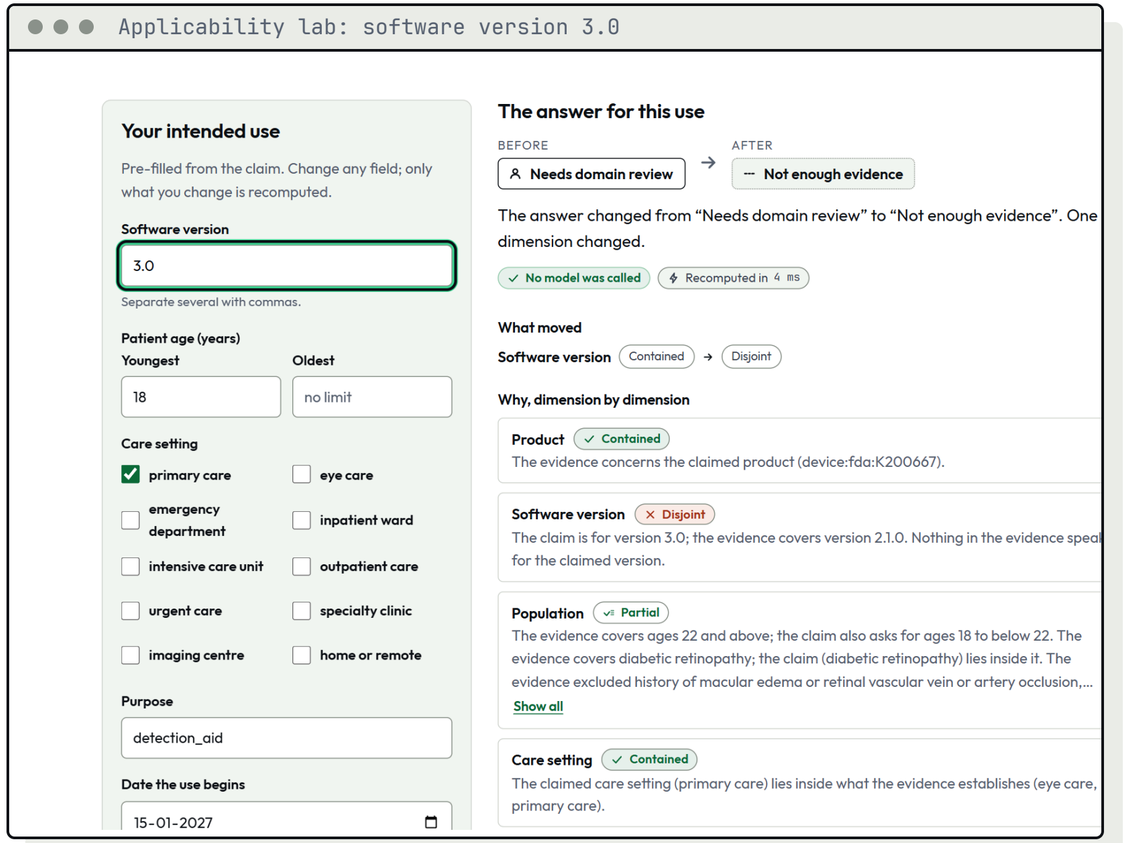
Sources: README.md · data/sample/run/labels.json · tests/engines/scout/RESULTS.md · +1

CLAIM X-RAY AND APPLICABILITY LAB

See the mismatch. Change the use.



Claim X-ray: the claim says 3.0, the evidence establishes 2.1.0. Each dimension opens to the exact passages.



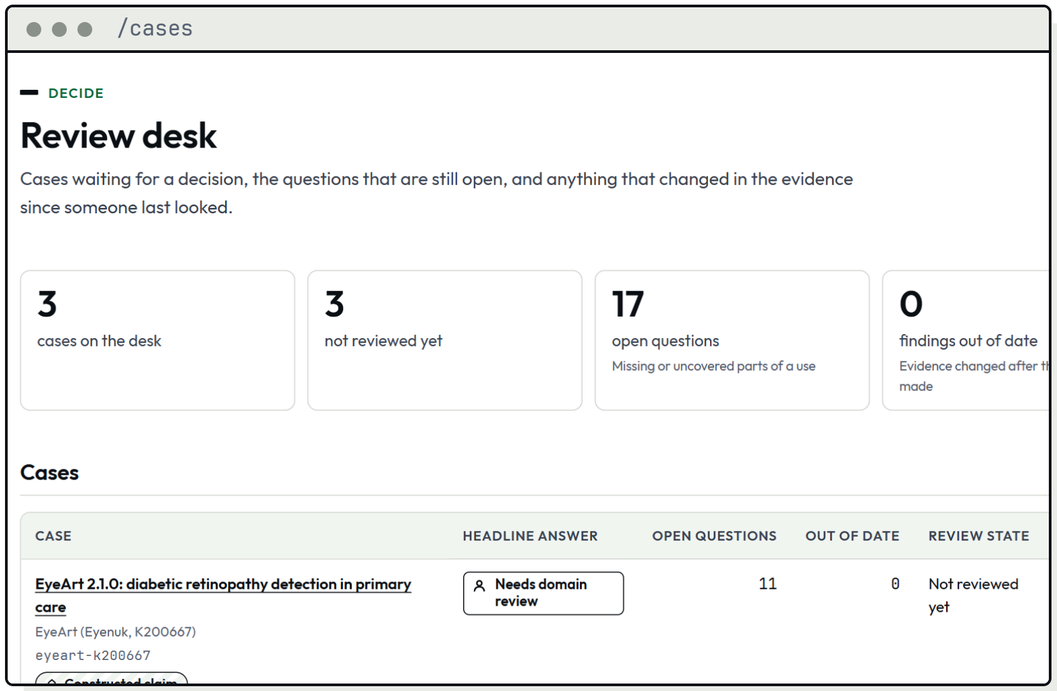
Applicability lab: version changed to 3.0. Answer recomputed, one dimension moved, no model called.

PRODUCT AND UX

A review workstation, not a chat box.

PRODUCT AND UX, 10 POINTS

“How intuitive, accessible, and easy to use is the product for its intended users?”



Review desk: open questions per case

- 14 screens, among them the desk, sources, guard, and per case an overview, matrix, X-ray, lab, timeline, investigation, drills and dossier
- Words and an icon on every finding, never colour alone
- A run-mode banner on every screen: sample or live, recorded or fresh
- Light and dark themes; keyboard-only flows
- Constructed claims are marked constructed on screen

CHECKED BY MACHINE

axe in light, dark and phone layouts; keyboard-only flows; 360 to 1440 px; 200 percent zoom; copy lint

NOT DONE

No moderated session with a real reviewer.

Sources: docs/05-requirements-matrix.md

automated checks; no moderated user session

FAILURE BEHAVIOUR

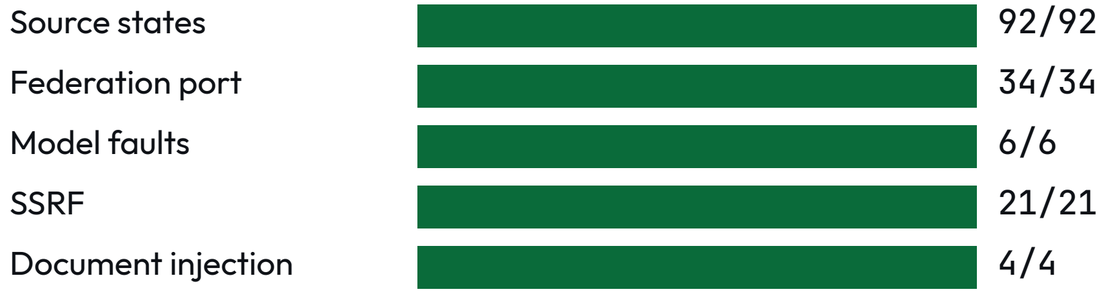
Seven drills. None turns a loss into support.

`source_off`	A named source becomes UNAVAILABLE
`source_empty`	A named source answers with no rows
`zetaris_off`	The federation engine is stopped
`truncated_result`	A federated result comes back cut short
`model_off`	The model route fails
`registry_changed`	A registry record changes after a finding
`injection_in_document`	A vendor document orders the system to mark a claim supported

THE INVARIANT THAT MUST HOLD

A lost required source never yields a supported label. Only findings that relied on a changed fact go stale. A model that is off, or an order hidden in a document, changes no label.

157 of 157 drills ended in the correct state



Real Zetaris API container stopped: the port reported **UNAVAILABLE after 4 ms** and, with the API up, the same port reports FORBIDDEN, so the two are never confused. The stack answered again **15,335 ms** after restart.

Not covered: the Nightshift route and the local lane were not faulted; concurrent users were not tried.

THE SIGNED DOSSIER

Sign the run. One changed character fails it.

THE EYEART DOSSIER HOLDS

11
Findings
with smallest mismatch and requests

101
Anchors
quoted passages, page and region

7
Source snapshots
content and request hashes

3
Federation statements
query id and result hash

14
Model replies
each with its receipt

180
Lineage events
run, job and dataset facets

WHAT THE SIGNATURE MEANS

The bundle has not changed since the key holder signed it. It does not show that any finding is clinically correct. A demo key does not identify a signer.



Export, verify, then change one character: the verifier names the failing item.

SPONSOR REQUIREMENTS

Each requirement, its code path, its evidence.

SPONSOR TECH, 10 POINTS

“How effectively does the solution use the sponsors' technology to address the problem and enhance the overall product?”

SPONSOR	ORGANIZERS ASK	WHAT IT DOES HERE	CODE	EVIDENCE	STATE
Zetaris	The data layer	One federated SQL over three silo types; four access identities; a Unified Semantic Layer with 7 data-quality rules	`src/federation/` `src/discovery/` `infra/zetaris/`	9 gates pass; natural-language route blocked	Working; one route not done
Meterless	One engine in a working feature	World Model marks dependent findings stale; H-MEM keeps confirmed links; Markovian reads long documents; Scout guards requests	`src/engines/` `third_party/meterless/`	Sponsor runners: H-MEM 13/13, World Model 8/8, Markovian 8/8	Working; Scout weak on new wording
Nightshift	The primary AI provider	Every recorded model reply came through Nightshift with a signed receipt	`src/models/nightshift.ts` `src/models/stack.ts`	61 of 61 receipts re-verified	Working; live fallback tested on doubles
NVIDIA Nemotron	The model the product runs on	Seven roles: Lightning 30B for volume, Ultra 550B for resolver and critic	`src/models/roles.ts`	output_examples/model-route.json	✓ Working
Cursor	Dev environment (optional per moderators)	Not used. Claude Code was the coding assistant	`none`	README.md, docs/provenance.md	✕ Not used
Bright Data	Only for the bounty track	Not applicable: primary track is Track 1	`none`	none	Not applicable

ZETARIS (1 OF 2)

The data layer: federation, policies, and what broke.

GATE	RESULT	WHAT THE EVIDENCE SHOWS
Z-HEAD	✓ Pass	One statement over three silos: 20 rows, equal to DuckDB cell by cell
Z-REST	✓ Pass	REST table over openFDA and ClinicalTrials.gov equals a direct call
Z-POL	✓ Pass, with two defects found	Four identities, four views; no confidential value reached the agent
Z-CAT	✓ Pass	Tags, schema drift visible through the catalog
Z-AUD	✓ Pass, with limits	Audit log lists 9 of 9 request ids; no SQL text
Z-TT	✓ Pass	VERSION AS OF returns the earlier registry state (Iceberg only)
Z-USL	✓ Pass	Six-table layer compiled and queried as a read-only identity
Z-PDF	✓ Pass	FORMAT PDF text for three FDA summaries
Z-NL	✕ Blocked	No natural-language-to-SQL service in the local build; three routes tried
Kill drill	✓ Pass	API stopped: UNAVAILABLE in milliseconds; engine restarts alone

115 SQL statements in the full run. Not run: Linux, macOS, concurrent users, Zetaris Cloud.

WHERE IT CHANGES AN OUTPUT

- Engine stopped: every required finding becomes not enough evidence
- Cross-source checks off: 16 of 48 mappable layouts held, 4 that must not register registered
- Four identities see different columns; 12 confidential values never reached the agent in 23 responses

DEFECTS WE FOUND AND REPORT

- Hide on a file-store table leaks the value under the next column's name
- A datasource refresh deletes every policy on that source
- A non-admin needs a REST call with no SQL form to read a USL

ZETARIS (2 OF 2)

The layer saved no run time. It saved code.

Same question, same 28 rows, median of nine rounds



Seven reads, one per silo table, joined in code

7

statements sent

48 lines of TypeScript

One hand-written federated statement

1

statements sent

about 40 lines of formatted SQL (the Evidence Matrix join)

One read of the layer

1

statements sent

4 lines

Ten tables, 89 columns, queried as a read-only identity. Engineering hours were not measured and no hours-saved figure is claimed.

Seven data-quality rules, run with RUN DQ

RULE	FAILS ON THE RECORDED STACK
collection_complete	0 of 7 fail
study_has_eligibility	143 of 157 fail
version_is_listed	0 of 20
claimed_age_plausible	0 of 18
claim_cites_authorization	0 of 18
case_vocabulary_defined	0 of 7
metric_proportion_bounds	0 of 75

The 143 failures are real: search summaries with no eligibility block. A deliberate truncation, 110 records cut to 39, failed the completeness rule and blocked that collection.

Sources: docs/zetaris-usl-reflection.md · docs/05-requirements-matrix.md

one machine, one question; ratios are not a benchmark

METERLESS: FOUR ENGINES

Each engine has a role, a result, and a caveat.

H-MEM

conformance 13/13

Keeps reviewer-confirmed links, accepted mappings and precedents.

- Repeated questions asked: 22 with no memory, 10 with scoped memory
- Wrong record reused: 4 without an identifier guard, 0 with it
- Unscoped control: 9 answers from the other hospital's records

`constructed session, 34 questions` `no interval`

World Model

conformance 8/8

Bi-temporal facts; marks exactly the dependent findings stale.

- 9 findings recomputed instead of 36, none missed
- No-engine arm: 7 stale unflagged, by construction
- TimeShift: stale set equals dependent set in 27 of 27

`by construction in part` `constructed session`

Markovian

conformance 8/8

Reads long documents in bounded chunks with a carried-over ledger.

- Planted restriction found: 3/4 (ids carryover), 2/4 (one full-length prompt), 0/4 (truncated prompt)
- Values carryover: 1/4
- 40-page constructed documents; four of them

`one document is not evidence`

Scout Intent

32-example harness exits 0

Guards every request and document-derived instruction; signs export contracts.

- Attacks through: 18/126 against 105/126 with no injection layer
- Wording written after the rules froze: recall 9/24 with rules alone
- Model classifier measured; off in the product

`held-out gates not met`

Sources: tests/engines/ABLATIONS.md · tests/engines/markovian/MA3-RESULTS.md ·… ablations on constructed inputs; null and by-construction results marked

NVIDIA NEMOTRON, THROUGH NIGHTSHIFT

Seven roles propose. Every reply is receipted.

ROLE	MODEL	WHY
planner	Lightning 30B	high volume, short structured plans
discoverer	Lightning 30B	mapping proposals from a bounded sample; a deterministic validator follows
extractor	Lightning 30B	highest volume: one call per document chunk; quote-exists is checked in code
support checker	Lightning 30B	semantic-support check in a separate context that sees only the passage and the assertion
reporter	Lightning 30B	prose from verified findings only; code rejects sentences without a finding id
resolver	Ultra 550B	identity adjudication inside the uncertain band; low volume, high consequence
critic	Ultra 550B	tries to break each draft finding; a stronger and different model than the one that drafted it

61 of 61

recorded replies with a signed Nightshift receipt, re-verified

ANSWERED BY

Ultra 550B: **29** Lightning 30B: **32**

REPLAYED RUN, REPLIES RECEIPTED

eyeart **14 of 14** · lunit **23 of 23** · transpara **23 of 23**

WHAT A RECEIPT DOES NOT SHOW

Where a reply was routed and which model answered, not that it is true. It carries no hash of the reply. NVIDIA's own API answers only if Nightshift is unreachable, and then the reply is labelled unreceipted. The comparison arms called Lightning through NVIDIA directly.

Sources: src/models/roles.ts · output_examples/model-route.json · README.md · +1

recorded replies; receipts show routing, not truth

SAFETY AND SECURITY

Fetched text is data. The guard's weak number is on the slide.

SSRF-safe fetch

- https only, GET only, no credentials forwarded
- Resolved addresses checked against private, loopback, link-local and cloud-metadata ranges
- DNS pinned; every redirect re-checked; size and time capped

21 of 21 SSRF drills correct

Untrusted text and the guard

Injection recall, rules only, as the product runs it



A model classifier raises the last row but pushes false blocks over the gate, so it is off (D-030).

Data classes and egress

- Public records may reach a model
- Vendor-confidential content: a local lane, or recorded consent
- Patient data never enters; none exists in the repository

17 egress decisions recorded in the EyeArt dossier; in the recorded run nothing left the machine.

Masking and receipts

- Zetaris policies mask commercial vendor columns for identities not assigned to a case
- Dossier signature: integrity, not truth
- Receipts: routing, not truth

Not verified: that a masked value cannot reach a model prompt end to end.

Sources: tests/engines/scout/RESULTS.md · evals/results/failureshift.json · docs/02-architect… guard figures are for rules alone, as the product runs it

HOW IT WAS EVALUATED

Five suites. The class of every result is printed beside it.

SUITE	TESTS	SIZE	CLASS	RESULT
DiscoveryShift	unknown schemas and meaning changes	102 layouts, 4 base tables	`consistency, one labeller`	first run 39/48 mapped, 44/44 caught
IdentityShift	similar devices, one true match	144 pairs, 14 families	`builder-labelled, dev set`	0 of 60 false merges
ScopeShift (A)	one dimension changed at a time	222 cases, 17 families	`consistency`	218 of 222 agree
ScopeShift (B)	authentic claims, same inputs for all arms	26 dev + 20 sealed, 12 + 5 devices	`builder-labelled`	next slide
TimeShift	a source changes after a finding	27 cases, 5 devices	`consistency`	stale set equals dependent set, 27 of 27
FailureShift	faults through the real ports	157 drills	`consistency`	157 of 157 correct state

THREE CLASSES

- Consistency: code against a rule table its authors wrote
- Builder-labelled: one builder's answer key
- Domain-validity: needs blind clinicians. None exists, so this deck has none

SEALED HELD-OUT SET

20 claims, 5 devices, sealed by file hash before any system ran: claims `a7e06def`, key `fcd1327e`. The extraction layer was not blind to those documents, so this is a second look, not a blind test.

ERRATUM LOG: 7 ENTRIES

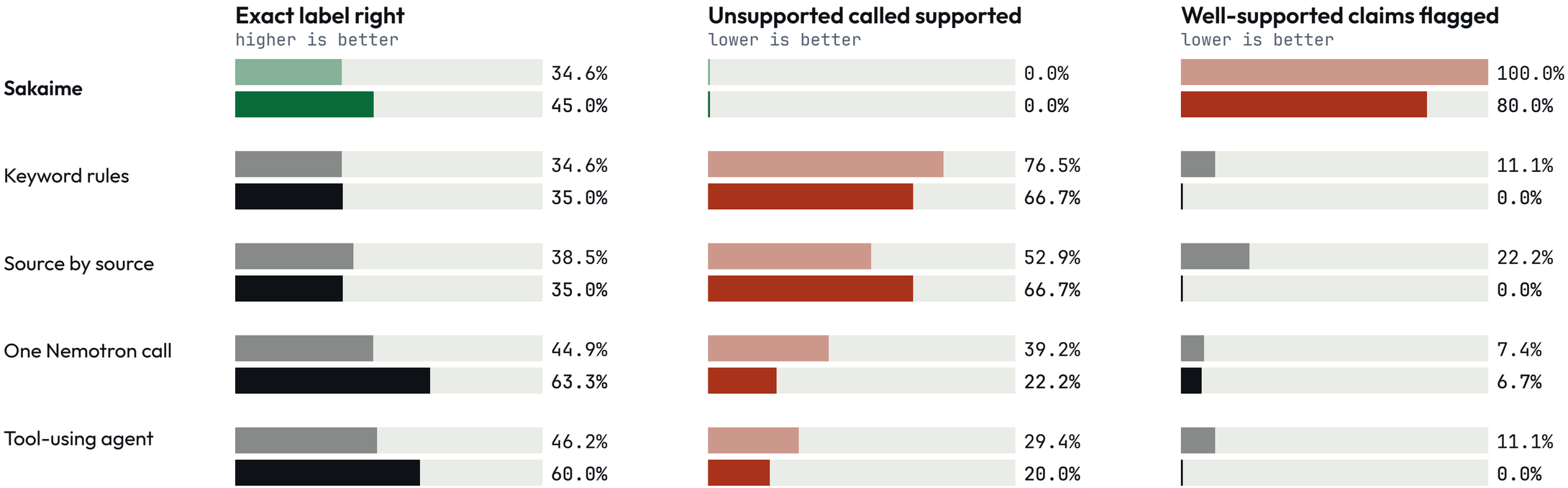
E-001: the Gate 0 key was edited after the builder saw a dry run of the compiler, so all 26 Gate 0 claims count as development. E-005: changes followed the first held-out score.

Sources: evals/RESULTS.md · evals/ERRATA.md · evals/SEALED.md · +5

every number links to a row of a results file

RESULTS, EVERY ROW

It never over-claims. It is less often exactly right.



development, 26 claims · held-out, 20 claims — Generic Nemotron arms: means over 3 runs (one call) and 1 run (agent).

Exact label: 9 of 26 dev, 9 of 20 held-out. Below both generic arms.

Unsupported called supported: 0 of 17 and 0 of 15. Generic arms 20.0% to 39.2%; keyword and source-by-source baselines 52.9% to 76.5%.

The cost: well-supported claims flagged 9 of 9 dev and 4 of 5 held-out. It does not over-claim; it also under-confirms.

Sources: evals/RESULTS.md · evals/results/scopeshift-b-dev.json · evals… builder-labelled; ScopeShift (B); constructed claims; held-out is a second look

ABLATIONS

Remove each part. Report what moved.

COMPONENT REMOVED	RESULT	STATUS
Compiler off (generic Nemotron arms)	exact 44.9% and 46.2% against 34.6%; they call 39.2% and 29.4% of unsupported claims supported, Sakaime none	measured
Compiler on complete hand-written evidence	exact 61.5% (16 of 26); the items encode the key's vocabulary	upper bound
Identity layer off	exact 9 of 26 against 9 of 26; effect shown in IdentityShift: false merges 0 of 60 against 23 of 60	null result
Cross-source checks off	16 of 48 mappable layouts held for lack of corroboration; 4 that must not register registered	measured
Model off, in discovery	34 of 48 mapped; 14 wrong mappings	measured
World Model off	7 stale findings unflagged by definition; 9 recomputed instead of 36 with it	by construction
H-MEM off	22 repeated questions asked; 10 with scoped memory (constructed session)	measured
Markovian: one full prompt, or truncated	restriction found 3/4 against 2/4 and 0/4; four documents	measured
Scout injection layer off	105/126 attacks through against 18/126	measured
Critic role; Nightshift receipts; federation off as an accuracy test	no label changed by the critic in the traces; the other two were not run	not measured

Sources: evals/RESULTS.md · data/discovery-shift/RESULTS.md · tests/engines/ABLATION…

pre-registered components; null and by-construction results marked

RUN IT YOURSELF

One command re-checks the claims offline, with no key.

DEMO, 15 POINTS

“How clearly does the demo show the solution, its functionality, and the value it provides to its intended users?”

```
# Node 22.19 or newer
git clone https://github.com/Pratiikpy/sakaime.git
cd sakaime
npm ci
npm run judge
npm run ui:build
npm run serve
```

67
lines PASS, 0 FAIL, working tree

5.5 s
seconds, no key, no network

WHAT JUDGE DOES

- Verifies every dossier: hashes, signature, quotes in the stored records
- Changes one byte and requires the verifier to fail
- Re-runs the compiler on the recorded inputs and compares every label
- Replays the source-off drills: never a supported label

SAMPLE_MODE=TRUE

Every external read is replayed from recordings of 2026-10-27. The compiler, identity scorer, verifiers and guard run for real. A question with no recording abstains visibly. A banner on every screen says which mode is on.

NOT DONE

A clean-clone run on Windows and Ubuntu. Live mode needs Docker for Zetaris and a model route.

Sources: README.md · .env.example · docs/05-requirements-matrix.md · +1

judge run in the working tree; clean clone not run

SUBMISSION COMPONENTS

What the organizers ask for, and where each one is.

COMPONENT	WHAT IS ASKED	WHERE	STATE
Demo video, 1 to 4 minutes	Main workflow, how the agent solves the challenge, key features, Zetaris and Meterless	`separate file; follows the screens in this deck`	Separate artefact
Slide deck	Problem and users, solution, workflow, stack, sponsor use, what was unique, visuals, future work	`this document`	✓ Here
GitHub repository and README	All source, setup, usage, dependencies; Colab links if used	`README.md; Apache-2.0; no Colab`	✓ Present
Sponsor technology explanation	Zetaris, Meterless, Cursor, NVIDIA, what was unique	`slides 23 to 27`	Here; no separate text
Sample inputs, outputs and traces	What goes in, what comes out, how the agents worked	`input_examples/, output_examples/ (traces/*.jsonl)`	✓ Present
SAMPLE_MODE fallback	Runs if an external service is down	`.env.example, recorded cassettes`	✓ Present
No secrets	.env ignored; .env.example holds placeholders	`.gitignore, .env.example`	✓ Present
Project description (HackOS form)	150 to 2000 characters: problem, mechanism, results with limits	`written for the form, not part of this deck`	Outside the deck
Pre-existing components declared	Product code and prepared data	`README, docs/provenance.md, metadata.json`	✓ Declared
Cursor as the development environment	Required in the organizers' text; optional per moderators	`not used`	✕ Not done
A test of a clean setup	Run it from scratch before submitting	`judge passes in the working tree`	✕ Not done

Sources: research/hackathon/organizer/01-welcome-guide.md · research/hackathon/organizer/04-approach-guide-sol… what the organizers ask, and where it is

LIMITS AND WHAT IS NOT VERIFIED

What this deck does not show.

No clinical validation
Rules cite published concepts; the logic and thresholds are our inference. 21 of 38 candidate rules are not implemented.

No buyer evidence
Nobody who runs such a review was interviewed; no clinician or independent reviewer read any key.

One reader
One builder wrote every answer key. The Gate 0 key was edited after a compiler dry run (E-001).

Few authentic mismatches
In the public data only 1 of 12 devices has a study stating the version the regulator reviewed.

Extraction gaps
Population is unknown in 22 of 26 development claims; stacked-table results are not read, so no numeric performance claim is supported today.

Constructed claims
Every vendor claim and hospital requirement is constructed and labelled so. No patient data.

Recorded replay
Sample mode replays recorded replies. Live mode was exercised once end to end.

Weak guard on new wording
Injection recall 9/24 on wording written after the rules froze; the classifier is off.

Receipts
They show routing, not truth, and carry no hash of the reply.

Zetaris
Natural-language route blocked; the layer saved no run time; a masked value reaching a prompt is not verified end to end.

Not run
Clean clone on Windows and Ubuntu; Constat and HealthIT on held-out claims; cost and latency; the critic ablation.

Not used
Cursor. Bright Data and Colab are not applicable.

Sources: README.md · evals/RESULTS.md · docs/DECISIONS.md · +2

NOT VERIFIED means no evidence either way

PRE-EXISTING COMPONENTS AND PROVENANCE

What existed before, what was studied, what was reused.

IDEAS STUDIED AND REBUILT IN NEW CODE

SOURCE	LICENCE
Splink (Ministry of Justice)	MIT
Docling (IBM)	MIT
OpenLineage	Apache-2.0
Airbyte protocol	MIT
Trino connector SPI	Apache-2.0
Graphiti	Apache-2.0

Plus PROBAST+AI, STARD-AI, TRIPOD+AI, IMDRF N41 and FDA PCCP guidance, as concepts. The executable rules are our inference.

VENDORED, UNMODIFIED

Meterless reference engines and conformance runners, Apache-2.0, commit `0aa7417`, file hashes checked by a test.

FROM AN EARLIER SUBMISSION

Design experience only: two sponsor request shapes, rebuilt and tested here. Font files (SIL OFL) copied. No code, copy or screenshots.

DATA

openFDA (CC0), ClinicalTrials.gov, PubMed, Europe PMC, FDA summaries: small hashed recordings. Vendor claims, requirements and the hospital database are constructed. No patient data.

AI ASSISTANCE

Code, tests and documents were written with Claude Code under the owner's direction. Cursor was not used.

ROADMAP, TEAM AND LINKS

What would change what we can honestly say.

1. **Independent reviewers label a held-out set blind**
   turns builder-labelled into domain validity
2. **Interview hospital AI intake committees**
   demand is not measured
3. **Read stacked tables; carry study population and setting**
   the main reason the pipeline is conservative
4. **Implement more compiler rules as the schema grows**
   21 of 38 are not implemented
5. **Zetaris natural-language route, when a build has one**
   blocked in the local build
6. **Domain packs: the same engine on addenda or tariff revisions**
   a thesis, not a claim

TEAM AND TOOLS

Solo build: **Prateek** (Pratiikpy). Code, tests and documents written with Claude Code under the owner's direction. Cursor not used.

LINKS

`github.com/Pratiikpy/sakaime`

`README.md` start here
`evals/RESULTS.md` every result
`docs/00-INDEX.md` design and decisions
`docs/provenance.md` what was reused
`npm run judge` re-check offline

Know where the evidence stops.

Sources: docs/DECISIONS.md · README.md · metadata.json · +1